

Say this first

A clinician pastes the note they already wrote. Aperta turns it into what the patient still understands at home: where on their body, one plain sentence, each medicine and how to take it, what to do next, and how to actually do it. In their language, read aloud, printed on one sheet. Then the clinician turns the screen around.

The three sentences that carry it

1. **The moment.** The most valuable screen in medicine is the one a doctor turns toward a patient, and nobody owns it. A rep is never in that room.
2. **The problem.** Patients forget most of a visit. The fix everyone agrees on is teach-back, and fifteen minutes does not allow it, so the explanation has to become an artifact.
3. **The discipline.** Nothing clinical is generated. A curated library decides every sentence, every skipped line is shown, and it runs with no API keys.

The question that always comes: how do you know it will not tell a patient something wrong?

Nothing clinical is written by a model. The 205 plain sentences are ours and fixed. What a medicine is for comes from 95 curated class rules keyed on drug classification, not on the drug's name. Warnings are quoted from the FDA label with a link back. A model, when configured, only translates or fills a parsing gap, and the demo runs with none. And the clinician reads every sentence before the screen turns, because turning it is a physical act they perform.

Numbers, all of them

205	conditions, each with a plain sentence, a drawing, the words a real note uses, ICD-10 prefixes
21	categories a patient would look under
18	hand-drawn anatomical views
9	step-by-step walkthroughs
95	curated drug-class rules
8	note formats in the regression suite
5	languages, Arabic right to left
0	API keys needed, published site included

Stack: Next.js 16, TypeScript, Tailwind v4, static export on GitHub Pages. RxNorm and RxClass for drug identity and class, openFDA for labels and packaging, MyMemory for translation, ElevenLabs for the example notes and the browser voice for everything else. Live at aperta.health.

Paste the wisdom-tooth note. It is the one that produces a walkthrough.

0:00	Paste the note. "This is the note the surgeon already wrote. Nobody types anything new."
0:08	The spot searches the body and lands on the jaw. "While it reads, the spot is looking for the place."
0:20	Point at Not used . "It refused to read that line as a drug. Refusing is a feature, and the clinician sees what was refused."
0:32	Before you prescribe , pick a medicine. "Same screen as the list. Duplicates, interactions, the boxed warning, quoted from the FDA label with a link back."
0:45	Press the button. Turn the laptop. Say nothing for two seconds.
0:52	Body, jaw, back molars, then the medicine screen. "One idea per screen, read aloud, advancing when the reading finishes."
1:05	The socket walkthrough. "This is the phone call three days later that nobody has to make."
1:18	Switch to Spanish . "Drug names are never translated, because the patient has to match them to the bottle."
1:26	Print this. "One sheet for the fridge."

Controls

Space pauses. Arrows move. The strokes along the top jump to a screen. Tap the picture to zoom out. Evergreen is the dark mode, on only by the button. The library is at /diagrams. Clean restart: `aperta.health/?note=wisdom&turn=1`, plus `&lang=Spanish` or `&slide=3`.

If something breaks

WHAT YOU SEE	WHAT TO DO
Wifi gone or slow	Use the two example notes. Prebuilt, including audio and Spanish. They need nothing.
No voice on a pasted note	Expected. Recorded audio ships only for the examples. Say so.
A check section missing	A public API was throttled. It reports what it checked. Re-run on another medicine.
Card looks thin	Show Not used. A thin card with a visible refusal is the better demo.
Page in a weird state	Reload the deep link above.

How it works, if they ask for the pipeline

Parse line by line, about eighty header aliases, prose scan when there are no headers. **Refuse** anything under Medications that does not read as a drug order, and show it. **Resolve** each medicine through RxNorm to its ingredients and classes, keeping the match only if the printed name shares a word or the line carried a dose. **Match** the diagnosis to the library, code before phrase, longest phrase wins. **Build** purposes from curated class rules, rewrite the sig, attach walkthroughs. **Deliver**: the clinician edits, then turns the screen.

Why is this engagement and not a patient app?

A patient never opens it. The clinician opens it, in the room, from the note they already wrote. Patient education is approved-content territory and adherence is the shared interest.

Who pays?

A manufacturer or a platform pays to have approved patient material delivered at the second a clinician needs it. A spacer walkthrough, a pen walkthrough, a plain purpose sentence. Placement in the explanation, not an impression.

Why would a busy doctor use it?

It takes the note they already wrote and gives back something they would otherwise say twice. A paste and a button.

How is it measurable if nothing is stored?

Nothing about the patient is stored. What is countable without an identifier is the material: which walkthrough, which language, which drug class. Not built yet, and we say so.

The EHR already prints instructions.

That printout is the note reformatted. No picture of where it is, no reading level, no voice, no language.

Why not ChatGPT?

A general model always answers, and here a confident wrong sentence is the harm. "Take the headache powder" is a real thing a fuzzy matcher gave us.

Is it regulated? A device?

As built it is patient education, labelled on every screen as a demo and not medical advice. It makes no recommendation and restates a diagnosis a clinician already made.

Who is responsible if a sentence is wrong?

The clinician, and the design makes that real. Every sentence is on their screen, editable, before the screen turns.

Can it hallucinate?

Not on the clinical path. No model writes any of it, and with no key the model never runs.

Off-label uses?

The real weakness of class-based purposes. There is a visit-context layer for cases we know, and every sentence is editable. A pharmacist should validate the library.

How do you know the spot is right?

Every spot was checked on a printed sheet beside its drawing. It is our anatomy, so the mark goes exactly where the injury is.

Does patient data leave the browser?

In English, only drug names, to the National Library of Medicine and the FDA. In another language the patient sentences go to a public translator, which is demo-grade and the first thing we would replace. Nothing is stored, ever.

What is deterministic?

Parsing, matching, resolution, purposes, sigs, walkthroughs and the whole prescriber check. Optional and absent here: a model for a thin parse, a photo, or translation.

How does the duplicate check work?

Set intersection over RxNorm ingredient identifiers, no strings, no model. Vicodin and Tylenol share one acetaminophen identifier, and the same identifier drives the daily ceiling.

A static site calling the FDA?

Those services take browser requests and need no key, so no server of ours is in the path. We inherit the anonymous rate limits.

How is it tested?

Eight note formats on every change: prose, an EHR export with codes, a cardiac discharge, a terse ED note, junk under Medications, a "Medicine:" header, both demo notes.

Past 205 conditions?

Growth is editorial, not technical. Generate candidates from a coded terminology, have a clinician approve the sentence.

Why hand-drawn, not 3D?

A photographic organ reads as gory after bad news, a third-party model cannot be marked with your fracture, and a drawing prints. We tried 3D and removed it.

Accessibility?

Text clears AA, keyboard throughout, descriptions on every drawing, reduced motion honoured, right to left for Arabic, everything read aloud and printable. Gaps: some hairlines under 3:1 and the custom dropdown's focus.

Who did what?

Lauren drew the anatomy, the walkthroughs and the portraits, and drove what a frightened person can take in. Aniketh built the parser, the library, the matcher and the prescriber check.

What was hardest?

Making it refuse. Easy to always answer, hard to decide in code that a line is not a drug and show that as care rather than failure.

Another week?

Teach-back, where the patient says it back and the tool checks the three facts survived. Then form-aware dosing, a pharmacist pass, and launching inside the chart.

All verified in our own code. Name it first, say why it is bounded, say the fix.

1. Dose sentences assume a pill

Puffs, drops and units are counted correctly but printed as "pill", so two puffs twice a day reads as "Two pills, twice a day". Patches, sprays and liquid volumes are not handled, and durations in weeks are dropped.

"Our dose sentences are pill-shaped today. It counts the dose right and names the form wrong, and that is next, because for an inhaler the form is the instruction."

2. Some medicine lines vanish instead of showing up under Not used

An order with no milligram-style strength, like "Albuterol HFA 2 puffs q4h PRN" or "Latanoprost 1 drop each eye", is discarded rather than listed. "Norco 5-325 mg" and "hydrocortisone 1% cream" fail the same way.

"The refusal is right, the reporting is not. Anything we cannot read should appear under Not used, and in that shape it does not. Small fix, and it matters, because silence is the one thing this project is not allowed to do."

3. Drug orders under "Plan" become diagnoses

"Assessment and Plan: 1. Type 2 diabetes, start metformin 500 mg BID" puts the whole line in the diagnosis list and finds no medicines.

"We tuned on discharge and after-visit summaries, where medicines have their own header. A Plan section needs the line split. We know the shape of the fix."

4. Negations and rule-outs are not guarded

"No history of stroke" and "rule out appendicitis" are still candidate diagnoses.

"Negation is the first thing a clinical parser needs, and we have a guard list rather than a guard rule. Known gap, known fix."

5. Concede these quickly if asked

Translation is free and unreviewed, and the shipped Spanish has at least one bad line. 205 conditions is small, and anything outside falls back to a whole-body drawing. Purposes are class-based, so an off-label use reads wrong until edited. Nothing has been validated clinically and no patient has used it. There is no analytics of any kind today.

Never say this

NOT THIS	THIS
"HIPAA compliant"	"It stores nothing, and here is exactly what is sent where."
"It diagnoses"	"It restates a diagnosis the clinician already made."
"The AI writes it"	"A curated library writes it. A model only translates or fills a parsing gap."
"It is accurate"	"It is deterministic, quoted from the label, and reviewed by the clinician."
"It is safe"	"It refuses rather than guesses, and it shows you what it refused."